16B 1 FUEL CELLS

SPECIFICATION REFERENCE
16.18 16.19

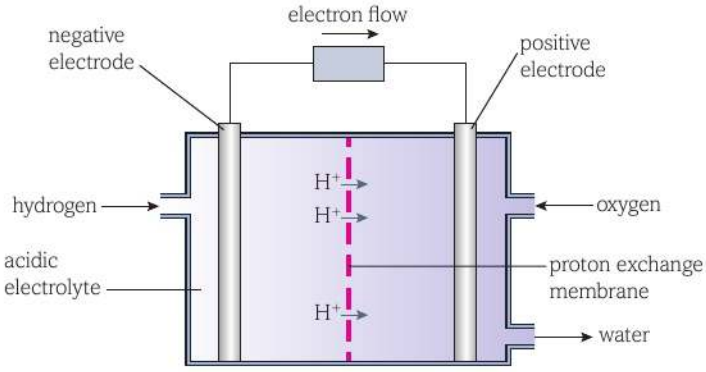
LEARNING OBJECTIVES

- Understand that fuel cells use the energy released on the reaction of a fuel with oxygen to generate a voltage.
- Know the electrode reactions that occur in a hydrogen–oxygen fuel cell.

HYDROGEN–OXYGEN FUEL CELL

A fuel cell produces a voltage from the chemical reaction of a fuel with oxygen. Several types of fuel cells have been developed, with the most common type using hydrogen as the fuel (the hydrogen–oxygen fuel cell). In this fuel cell, hydrogen is supplied externally as a gas, and the cell can operate as long as the fuel supply is maintained.

Other fuel cells use methanol/ethanol or hydrogen-rich methanol as the fuels.



▲ **fig A** A hydrogen–oxygen fuel cell.

In the hydrogen–oxygen fuel cell (**fig A**), both metal electrodes are coated with platinum, which catalyses the reactions that take place at the electrodes. The two reactions that take place in the presence of an acidic electrolyte are shown below.

At the negative electrode: $H_2(g) \rightarrow 2H^+(aq) + 2e^-$.

At the positive electrode: $\frac{1}{2}O_2(g) + 2H^+(aq) + 2e^- \rightarrow H_2O(l)$.

The hydrogen ions (protons) pass through the proton exchange membrane, which allows them to enter the compartment containing the positive electrode, where they can react with oxygen.

The overall cell reaction is:

$$H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(l)$$

There is also a hydrogen–oxygen fuel cell that has an alkaline electrolyte. The electrode reactions for this cell are shown below.

At the negative electrode: $H_2(g) + 2OH^-(aq) \rightarrow 2H_2O(l) + 2e^-$.

At the positive electrode: $\frac{1}{2}O_2(g) + H_2O(l) + 2e^- \rightarrow 2OH^-(aq)$.

The overall cell reaction is the same as above:

$$H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(l)$$

Advantages of hydrogen–oxygen fuel cells

- They offer an alternative to the direct use of fossil fuels such as petrol and diesel.
- Unlike fossil fuels, their products do not include carbon dioxide or pollutants such as carbon monoxide and the oxides of nitrogen.
- They are lighter and more efficient than engines that use fossil fuels.

Problems associated with hydrogen–oxygen fuel cells

Hydrogen explodes easily when ignited, so great care has to be taken when transporting it. We will consider the following options in turn.

- compressing the gas
- adsorbing it onto the surface of a suitable solid material
- absorbing it into a suitable material.

Compressing the gas has disadvantages in terms of safety. Transporting hydrogen under pressure is particularly hazardous.

Many metals adsorb hydrogen, but so far no firm conclusion has been reached as to the best one to use. One of the most promising areas of research is the use of carbon nanotubes to adsorb hydrogen.

Many metal hydrides absorb hydrogen, which can be released under the correct conditions. The problem is that high temperatures are required to release the hydrogen.

Another problem is the supply of hydrogen. At the moment, most of the hydrogen is produced from methane, which is a finite or limited source. It can be produced using renewable sources, but it is an expensive process. Integrated wind-to-hydrogen (power-to-gas) plants, using electrolysis of water, are exploring technologies to deliver costs low enough, and quantities great enough, to compete with traditional energy sources.

Fig B shows a prototype car that uses fuel cell technology.

DID YOU KNOW?

Fuel cells have been used in various kinds of vehicles, including forklifts, especially in indoor applications where their clean emissions are important to air quality, and in space applications. The first commercially produced hydrogen fuel cell automobile, the Toyota Mirai was introduced in 2015, after which Hyundai and Honda entered the market. Fuel cells are also being developed and tested in trucks, buses, boats, motorcycles and bicycles, among other kinds of vehicles.



▲ **fig B** A car that uses a hydrogen–oxygen fuel cell.

CHECKPOINT

1. Nickel–cadmium (NiCd) cells are used in many electronic devices because they are very reliable, economical and easy to use.

The standard electrode potentials for the redox systems in NiCd cells are:

$$Cd(OH)_2(s) + 2e^- \rightleftharpoons Cd(s) + 2OH^-(aq) \quad E^\ominus = -0.88\,V$$

$$NiO(OH)(s) + H_2O(l) + e^- \rightleftharpoons Ni(OH)_2(s) + OH^-(aq) \quad E^\ominus = +0.52\,V$$

(a) What is the standard cell potential, $E^\ominus_{cell}$, of a NiCd cell?

(b) Construct the overall reaction that takes place when a NiCd cell is producing an electric current.

(c) Use oxidation numbers to show which element is oxidised and which is reduced during the discharge of a NiCd cell.

2. Fuel cells are being developed as an alternative to the use of petrol and diesel in cars. The most common fuel used in these cells is hydrogen, but cells using other fuels are being developed.

In a methanol fuel cell, the fuel is supplied at the negative electrode, with oxygen at the positive electrode.

The ionic half-equation for the reaction of oxygen at the positive electrode is:

$$\tfrac{1}{2}O_2(g) + 2H^+(aq) + 2e^- \rightarrow H_2O(l)$$

(a) Write a chemical equation for the complete combustion of methanol in oxygen.

(b) Use the equation in (a), and the ionic half-equation for the reaction of oxygen at the positive electrode, to deduce an ionic half-equation for the reaction that takes place at the negative electrode.

SKILLS ETHICS, INNOVATION

(c) Give two advantages of using fuel cells compared with the combustion of petrol or diesel.

(d) Give an advantage of using methanol, rather than hydrogen, in a fuel cell for use in cars.

3. (a) What is the essential difference between a fuel cell and a conventional electrochemical cell?

SKILLS ETHICS, INNOVATION

(b) State two ways that hydrogen might be stored as a fuel for use in cars.

(c) Suggest why some scientists consider that using hydrogen as a fuel for cars uses up more energy than using petrol or diesel as the fuel.

16B 2 REDOX TITRATIONS

SPECIFICATION REFERENCE
16.15 16.16
CP13A CP13B

LEARNING OBJECTIVES

- Be able to carry out both structured and unstructured titration calculations involving redox reactions, including iron(II) ions and potassium manganate(VII) and sodium thiosulfate and iodine.
- Understand the methods used in redox titrations.

REDOX TITRATIONS WITH POTASSIUM MANGANATE(VII), $KMnO_4$

Potassium manganate(VII) is a powerful oxidising agent. It is used for the quantitative estimation of many reducing agents, especially compounds of iron(II) and for ethanedioic acid and its salts.

In acidic solution, the half-equation for the reduction of the manganate(VII) ion is:

$$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)$$

In alkaline solution, potassium manganate(VII) yields manganese(IV) oxide as a brown precipitate, and this interferes with the end point colour. For this reason, therefore, potassium manganate(VII) is almost always used in solutions that are sufficiently acidic to prevent the formation of manganese(IV) oxide.

As the titration proceeds, manganese(II) ions, Mn^{2+}, accumulate but, at the dilution used, they give a colourless solution. As soon as the potassium manganate(VII) is in excess, the solution becomes pink. It therefore acts as its own indicator – the end point is the first permanent pink colour.

TITRATION OF POTASSIUM MANGANATE(VII) WITH IRON(II) IONS

In this reaction, the iron(II) ions are oxidised and the manganate(VII) ions are reduced.

The ionic half-equations involved are:

$$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)$$
$$Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$$

The overall equation for the reaction is:

$$MnO_4^-(aq) + 8H^+(aq) + 5Fe^{2+}(aq) \rightarrow Mn^{2+}(aq) + 4H_2O(l) + 5Fe^{3+}(aq)$$

An aqueous solution containing MnO_4^- ions is purple. In reasonably concentrated solutions containing Fe^{2+}, Fe^{3+} and Mn^{2+}, the ions have the following colours:

$Fe^{2+}(aq)$ – pale green

$Fe^{3+}(aq)$ – yellow

$Mn^{2+}(aq)$ – pale pink

However, the solutions used in titrations are usually so dilute that they appear colourless.

The method of performing the titration is:

- Pipette an accurately measured volume, usually 25.0 cm^3, of the iron(II) solution into a conical flask and then add a small volume of dilute sulfuric acid.
- Slowly add potassium manganate(VII) solution of accurately known concentration from a burette and swirl the mixture.
- The potassium manganate(VII) solution will turn colourless until all of the iron(II) ions have been oxidised.
- The addition of one more drop of potassium manganate(VII) solution will turn the mixture pale pink (**fig A**).



▲ **fig A** End point colour in a potassium manganate(VII) titration.

EXAM HINT

Could acidified potassium dichromate be used as an oxidising agent for quantitative redox titrations? In theory, the answer is yes. However, in practice, the answer is no. This is because the colour change from orange to green has no clear end point.

WORKED EXAMPLE

25.0 cm^3 of iron(II) sulfate solution reacts with 22.40 cm^3 of 0.0200 mol dm^{-3} potassium manganate(VII) solution. Calculate the concentration of the iron(II) sulfate.

$5Fe^{2+}(aq) \equiv MnO_4^-(aq)$

25.0 cm^3 of x mol dm^{-3} $Fe^{2+} \equiv$ 22.40 cm^3 of 0.0200 mol dm^{-3} MnO_4^-

$$\frac{25.0x}{22.40 \times 0.0200} = \frac{5}{1}$$

$$x = \frac{5 \times 22.40 \times 0.0200}{25.0} = 0.0896$$

$[FeSO_4(aq)] = 0.0896$ mol dm^{-3}

TITRATION OF POTASSIUM MANGANATE(VII) WITH ETHANEDIOIC ACID

In this reaction, the ethanedioic acid is reduced to carbon dioxide. The ionic half-equation for this reduction is:

$$H_2C_2O_4(aq) \rightarrow 2H^+(aq) + 2CO_2(g) + 2e^-$$

The overall equation for the reaction is:

$$2MnO_4^-(aq) + 6H^+(aq) + 5H_2C_2O_4(aq) \rightarrow 2Mn^{2+}(aq) + 8H_2O(l) + 10CO_2(g)$$

Aqueous ethanedioic acid is pipetted into the conical flask (Care: ethanedioic acid is extremely poisonous, so a pipette filler *must* be used) and is then acidified with dilute sulfuric acid. Aqueous potassium manganate(VII) is added from the burette. However, the reaction is very slow, so the ethanedioic acid is heated to around 60 °C before starting the titration. Even then, the reaction is slow to begin with and the pink colour does not immediately disappear when the first sample of potassium manganate(VII) is added. This may lead you to erroneously think that the titration is complete. After the initial sample is added, the Mn^{2+} ions produced act as a catalyst and the reaction speeds up, allowing you to titrate normally and obtain an accurate end point.

Potassium manganate(VII) can also be used to estimate the amount of ethanedioate ion present in a solution. The equation for the reaction is:

$$2MnO_4^-(aq) + 16H^+(aq) + 5C_2O_4^{2-}(aq) \rightarrow 2Mn^{2+}(aq) + 8H_2O(l) + 10CO_2(g)$$

DID YOU KNOW?

A reaction in which a product acts as a catalyst is said to be autocatalysed. Autocatalysis is discussed in Topic 11.

LEARNING TIP

The dilute sulfuric acid is boiled to remove dissolved oxygen that might otherwise oxidise some of the iron(II) ions to iron(III).

EXAM HINT

In your exam you may be asked to **calculate** the measurement uncertainty for a titration. You can find out more about this in **Topic 8 (Book 1: IAS)**.

PRACTICAL SKILLS CP13A

A redox titration: the determination of the number of moles of water of crystallisation per mole of hydrated iron(II) sulfate crystals

Note that this suggested practical uses techniques and skills similar to those needed in the **Core Practical 13a: Redox titrations with iron (II) ions and potassium manganate (IV)** in your Lab Book.

Procedure

1 Bring to the boil 100 cm^3 of dilute sulfuric acid and then allow it to cool to room temperature.
2 Weigh accurately a weighing bottle containing approximately 7 g of hydrated iron(II) sulfate crystals.
3 Dissolve the crystals in the cool acid and reweigh the bottle. Transfer the solution to a 250 cm^3 volumetric flask. Add several washings from the beaker and then make up to the mark. Shake well.
4 Titrate a 25.0 cm^3 sample of the iron(II) solution, as before, with the standard potassium manganate(VII) solution.
5 Perform repeat titrations until you obtain concordant results.

Calculations

1 Calculate the mean titre from your concordant results.
2 Calculate the concentration of the iron(II) sulfate solution.
3 Calculate the mass of iron(II) sulfate in the 250 cm^3 of solution.
4 Calculate the mass of water in the mass of crystals taken and hence calculate the number of moles of water per mole of salt.

Sample results

Concentration of potassium manganate(VII) solution = 0.0198 mol dm^{-3}

Mass of hydrated iron(II) sulfate = 6.84 g Mean titre = 24.95 cm^3

[Molar masses: $FeSO_4$ = 151.9 g mol^{-1}; H_2O = 18.0 g mol^{-1}]

$$5FeSO_4 \equiv KMnO_4$$

25.0 cm^3 of x mol dm^{-3} $FeSO_4$ ≡ 24.95 cm^3 of 0.0198 mol dm^{-3} $KMnO_4$

$$\frac{25.0x}{24.95 \times 0.0198} = \frac{5}{1}$$

$[FeSO_4(aq)]$ = 0.0988 mol dm^{-3}

Mass of $FeSO_4$ in 250 cm^3

$$x = \frac{5 \times 24.95 \times 0.0198}{25.0} = 0.0988$$

Mass of water = (6.84 − 3.75) = 3.09 g

If the formula of the hydrated salt is $FeSO_4.yH_2O$, then

$$\text{Mass of } FeSO_4 \text{ in 250 cm}^3 = 0.0988 \times 151.9 \times \frac{250}{1000} = 3.75 \text{ g}$$

$$\frac{18.0y}{151.9} = \frac{3.09}{3.75}$$

$$y = \frac{3.09 \times 151.9}{18.0 \times 3.75}$$

$$= 6.95$$

∴ The formula of hydrated iron(II) sulfate is $FeSO_4.7H_2O$

! Safety Note: Wear eye protection and avoid skin contact with the solutions. Make sure the top of the burette is below eye level when you fill it.

REDOX TITRATIONS WITH IODINE AND SODIUM THIOSULFATE

Thiosulfate ions reduce iodine to iodide ions. The ionic half-equations involved are:

$$2S_2O_3^{2-}(aq) \rightarrow S_4O_6^{2-}(aq) + 2e^-$$
$$I_2(aq) + 2e^- \rightarrow 2I^-(aq)$$

The overall equation for the reaction is:

$$2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

The reaction can be used for the direct estimation of iodine, or for the estimation of a substance that can take part in a reaction that produces iodine.

The indicator used in titrating iodine with sodium thiosulfate is starch solution. With free iodine it produces a deep blue-black colour. The blue-black colour disappears as soon as sufficient sodium thiosulfate has been added to react with all of the iodine.

The procedure is as follows.

- Add sodium thiosulfate solution from the burette to the iodine solution until the original brown colour of the iodine changes to pale yellow.
- Add a few drops of starch solution to produce a blue colouration.
- Add the sodium thiosulfate solution drop by drop until the blue-black solution turns colourless.

If the starch is added too early, it adsorbs some of the iodine and reduces the accuracy of the titration.

If starch were not added, it would be very difficult to detect the end point. The colour of the iodine solution becomes very faint towards the end of the reaction, and this makes it difficult to accurately assess when the end point has been reached.

PRACTICAL SKILLS CP13B

A redox titration: determination of the percentage of copper(II) ions in a sample of hydrated copper(II) sulfate

Note that this suggested practical uses techniques and skills similar to those needed in the **Core Practical 13b: Redox titrations with sodium thiosulfate and iodine** in your Lab Book.

The copper(II) sulfate solution must be free from anything but a trace of mineral acid, otherwise the end point is not accurate. This solution produces iodine when potassium iodide is added. The ionic equation for this reaction is:

$$2Cu^{2+}(aq) + 4I^-(aq) \rightarrow 2CuI(s) + I_2(aq)$$

The iodine produced is then titrated with standard sodium thiosulfate solution.

Procedure

1 Weigh accurately a weighing bottle containing approximately 6 g of hydrated copper(II) sulfate crystals.

2 Transfer the crystals to a 250 cm^3 volumetric flask and reweigh the bottle.

3 Add sodium carbonate solution until a *slight* permanent bluish precipitate of copper(II) carbonate is formed.

4 Acidify the mixture with a little dilute ethanoic acid until a clear blue solution is formed. Make up the solution to 250 cm^3 with deionised (or distilled) water and shake well.

5 Pipette 25.0 cm^3 of the solution into a conical flask and add about 1.5 g of solid potassium iodide.

6 Titrate the iodine produced with sodium thiosulfate solution, adding starch solution just as the colour of the mixture changes to pale yellow. The end point is achieved when you can see a white suspension of insoluble copper(I) iodide, CuI(s), in a colourless solution.

7 Perform repeat titrations until you obtain concordant results.

Calculation

Calculate the percentage of copper(II) ions in the sample of hydrated copper(II) sulfate crystals.

Sample results

Concentration of sodium thiosulfate solution = 0.0995 mol dm^{-3}

Mass of hydrated copper(II) sulfate crystals = 5.85 g

Mean titre = 25.45 cm^3

[Molar mass of Cu^{2+} = 63.5 g mol^{-1}]

$$2Cu^{2+} \equiv I_2 \equiv 2Na_2S_2O_3$$
$$\therefore \quad Cu^{2+} \equiv Na_2S_2O_3$$

25.0 cm^3 of x mol dm^{-3} Cu^{2+} ≡ 23.45 cm^3 of 0.0995 mol dm^{-3} $Na_2S_2O_3$

$$\therefore \quad 25.0x = 23.45 \times 0.0995$$
$$\therefore \quad x = 0.0933$$

Solution = $0.0933 \text{ mol}^{-1} \text{ dm}^{-3} \times 63.5 \text{ g mol}^{-1} \times \frac{250}{1000} = 1.48$ g

Percentage of Cu^{2+} in the sample = $\frac{1.48}{5.85} \times 100 = 25.3\%$

! Safety Note: Wear eye protection and avoid skin contact with the solids and liquids, especially the hot acid. Boil the acid gently for a short time in a larger beaker to avoid it boiling over and to cool it faster. Make sure the top of the burette is below eye level when you fill it.

CHECKPOINT

1. An iron nail of mass 1.50 g was reacted with dilute sulfuric acid. The resulting solution was transferred to a volumetric flask and made up to 250 cm^3 with deionised water.

A 25.0 cm^3 sample of this solution reacted with exactly 24.40 cm^3 of 0.0218 mol dm^{-3} potassium manganate(VII) solution.

The ionic equation for the reaction between the iron nail and the acid is:

$$Fe(s) + 2H^+(aq) \rightarrow Fe^{2+}(aq) + H_2(g)$$

Calculate the percentage by mass of iron in the nail.

[Molar mass of iron = 55.8 g mol^{-1}]

2. A 50 cm^3 volume of a solution of hydrogen peroxide was diluted with water to 1.00 dm^3. A 25.0 cm^3 sample of this solution was acidified and then titrated against 0.0200 mol dm^{-3} potassium manganate(VII). The mean titre value was 23.90 cm^3. Calculate the concentration, in mol dm^{-3}, of the original solution of hydrogen peroxide.

The ionic half-equation for the oxidation of hydrogen peroxide is:

$$H_2O_2(aq) \rightarrow 2H^+(aq) + O_2(g) + 2e^-$$

3. Calculate the volume of a 0.0200 mol dm^{-3} solution of potassium manganate(VII) required to completely oxidise 1.00 g of iron(II) ethanedioate (FeC_2O_4).

[Molar mass of FeC_2O_4 = 143.8 g mol^{-1}]

4. An aqueous solution contains a mixture of iron(II) ions and iron(III) ions. A 25.0 cm^3 volume of the solution required 21.60 cm^3 of 0.0210 mol dm^{-3} potassium manganate(VII) for oxidation.

A separate 25.0 cm^3 sample of the solution was reacted with an excess of zinc amalgam in order to reduce the iron(III) ions to iron(II). This sample, after filtration to remove the excess zinc, required 44.40 cm^3 of 0.0210 mol dm^{-3} potassium manganate(VII) solution for complete oxidation.

Calculate the concentrations, in mol dm^{-3}, of both the iron(II) and iron(III) ions in the aqueous solution.

SKILLS PROBLEM-SOLVING CREATIVITY

5. When bleaching powder reacts with a dilute acid, chlorine is produced. This chlorine is available for bleaching and is known as 'available' chlorine.

A 2.50 g sample of bleaching powder is added to an excess of potassium iodide solution and the mixture is then acidified. The solution is placed in a volumetric flask and made up to 250 cm^3 with deionised water. A 25.0 cm^3 volume of this solution required 23.20 cm^3 of 0.105 mol dm^{-3} sodium thiosulfate solution to oxidise the iodine present.

Calculate the percentage of 'available' chlorine in the 2.50 g sample of bleaching powder.

$$Cl_2(aq) + 2I^-(aq) \rightarrow 2Cl^-(aq) + I_2(aq)$$

[Molar mass of Cl_2 = 71.0 g mol^{-1}]

6. A 25.0 cm^3 sample of of 0.0210 mol dm^{-3} potassium peroxydisulfate, $K_2S_2O_8$, was treated with excess potassium iodide. The iodine produced reacted with 21.00 cm^3 of 0.0500 mol dm^{-3} sodium thiosulfate solution.

Suggest a likely ionic equation for the reaction between $S_2O_8^{2-}$ ions and I^- ions.